Sahaj Bhadja

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Summary

Frontend software engineer with 2+ years shipping responsive production applications, reusable interfaces, and API-driven workflows. Skilled in TypeScript, React, Next.js, React Native, Node.js, performance optimization, testing, and iterative product delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected multi-tenant product workflows on Azure SQL and Blob Storage for 500+ users, applying row-level security, normalized schemas, and indexed queries to reduce customer-facing dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant asynchronous workflows for 1,000+ monthly jobs with explicit state transitions, retries, failure isolation, and production monitoring, improving throughput by 3x while preventing cascading user-facing failures.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered reusable React workflows and Flask REST services handling 10,000+ daily requests, integrating secure uploads, AWS RDS persistence, and responsive state updates while maintaining stable performance during peak usage.
- Implemented concurrent processing for 1,000+ files with validation, error isolation, structured logging, code reviews, and automated tests, reducing product feedback turnaround by 60% for 150+ instructors.

USC AD Labs **Los Angeles, CA**
Machine Learning Researcher *January 2025 – May 2025*
- Built Python application pipelines across 100,000+ trajectory records, producing reproducible metrics, logs, and visual summaries that improved debugging, benchmarking, and experiment traceability for interactive system analysis.
- Documented application behavior, edge cases, and validation outputs for researchers, translating complex system signals into actionable findings while maintaining testable procedures and clear technical artifacts for team review.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Engineered cross-platform services with object-oriented design, Jenkins pipelines, automated builds, and 20+ PyTest and Mockito suites across Windows, Linux, and macOS, strengthening release quality and application maintainability.
- Optimized API and persistence workflows using AWS Glue, Amazon S3, SQL, indexing, caching, and pagination, improving application response times by 30% and production pipeline efficiency by 68%.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed responsive MERN interfaces with REST APIs and WebSockets, synchronizing 2M+ records across modular services while reducing conflicts through validation controls, fault handling, and real-time state updates.
- Automated Docker-based builds and deployment pipelines using Jenkins CI/CD, unit validation, and release checks, reducing deployment time by 50% while improving repeatability across development and production environments.

Skills

Frontend: TypeScript, JavaScript, React, Next.js, React Native, HTML5, CSS3, TailwindCSS, responsive design
UI Engineering: reusable components, state management, mobile interfaces, accessibility, performance optimization, user-centered design
Backend & Data: Node.js, Express.js, REST APIs, Flask, PostgreSQL, MySQL, MongoDB, Firebase, AWS RDS
Testing & Delivery: unit testing, integration testing, PyTest, Mockito, debugging, Git, code reviews, CI/CD, Jenkins
Product Practices: Agile, cross-functional collaboration, requirements analysis, technical documentation, iterative delivery, API integration, AI-assisted workflows

Projects

PiXMod | *Next.js, TypeScript, Angular, MongoDB, Stripe, AI APIs*
- Built a responsive AI-powered image workspace with search, segmentation, editing, and subscription workflows, processing 1,000+ assets while increasing premium conversion by 18% through streamlined product interactions.

FinNexus | *Next.js, TypeScript, MongoDB, WebSockets, Plaid*
- Developed responsive transaction interfaces supporting 50,000+ users and 1M+ records, integrating secure APIs, real-time state synchronization, persistent filtering, and token-based access controls across scalable application workflows.

HMIS | *React, TypeScript, Appwrite, REST APIs, TailwindCSS*
- Created user-centered appointment and patient-history interfaces with authenticated persistence and API integrations, enabling clinicians to access records 2x faster and coordinate 100+ appointments through responsive application workflows.